

A smaller world. A mountain that stays immense.

Curiosity's recorded journey, surrounded by open country. A proposed Gale footprint for review—not a constructed level.

01 / THE RECORDED JOURNEY

One open region around the real route



Gold: NASA's recorded traverse. **Mint:** proposed free-roam envelope. North is up. The map preserves real geography; no distance compression is applied here. Numbered pins are dated drive references, not proposed missions.

9 × 17 km

Proposed geographic envelope · 153 km²

Sol 5021

Latest waypoint in the downloaded NASA snapshot

38.38 km

Rover odometer · not straight-line distance or game time

Trace Curiosity's progress

In the interactive version, move the slider or select a numbered pin. The white ring locates a recorded drive endpoint; the complete gold route stays visible.

Room to leave the trail

The full recorded traverse fits inside this broad rectangle. At its extremes, the route has approximately 1.5–1.7 km of surrounding ground before the proposed edge. Players may explore between and beyond its bends wherever slopes permit.

This is an outer planning envelope. Cliffs, steep slopes and sandy terrain will reduce the usable area; it does not promise that every square metre is driveable.

What this establishes

- All downloaded route segments are included, through sol 5021.
- Landing-basin ground connects to the lower mountain ascent.
- Mountain terrain continues beyond the southern and eastern edges.
- Existing Gale arrival composition remains a continuity reference; its local coordinates still need registration to this terrain.

What is measured, and what remains proposed

Dated route references

Pin	Mars day	Odometer	Above first waypoint
1	Sol 3	0.00 km	+0 m
2	Sol 504	4.65 km	+11 m
3	Sol 997	10.82 km	+54 m
4	Sol 1500	14.76 km	+142 m
5	Sol 1999	18.68 km	+341 m
6	Sol 2481	21.32 km	+368 m
7	Sol 3000	24.19 km	+414 m
8	Sol 3504	28.42 km	+588 m
9	Sol 3974	31.36 km	+737 m
10	Sol 4500	34.62 km	+812 m
11	Sol 5021	38.38 km	+1021 m

Elevations come from NASA's waypoint `elev_geoid` values, not the shaded overview. Sols are Mars days. Pins nearest each 500-sol interval are chronological references; no landmark identity is inferred from a date.

Route integrity

1,312 source features / 1,316 independent line parts / 26,956 vertices. Summed mapped lines: 38.050 km; endpoint odometer: 38.384 km. Adjacent source-part gaps total 333.6 m (largest 102.1 m). These different measures are not interchangeable; gaps remain unjoined.

The original NASA files are preserved alongside this survey. Their independent line parts are drawn without artificial connecting chords. No proposed future traverse is added. Retrieval date identifies the snapshot, not a live rover position.

Survey proposal only · No gameplay, controls, terrain, saves or campaign changes.

Terrain evidence

The background is sampled from the USGS *MSL Gale merged elevation mosaic v3*, built from HRSC, CTX and HiRISE terrain tied to MOLA. Its 1 m output grid does not mean uniform 1 m source detail or accuracy. This survey uses one native sample per approximately 100 m cell; it is adequate for regional context, not small obstacles or driveability.

The sampled context is 97.60% valid elevation cells. Coverage and fine source quality still need checking at individual landmarks. This overview does not certify slopes, rover clearance or exact photograph viewpoints.

Coordinates use planetocentric latitude, east-positive longitude, a 3,396,190 m Mars sphere, and a local east/north frame centred at NASA's first waypoint (137.441633° E, 4.589467° S). East distances include the cosine of that latitude. The same transform registers route, boundary and terrain.

Primary sources

- [NASA: Curiosity's route and map explanation](#)
- [NASA recorded traverse data](#)
- [NASA drive endpoints and odometer](#)
- [USGS elevation mosaic and provenance](#)
- [Pinned-source record and checksums](#)
- [Survey measurements](#)

Route: NASA/JPL-Caltech. Terrain: USGS Astrogeology, NASA/JPL-Caltech, ESA/DLR and University of Arizona. Shading is generated from measured elevations, not an orbital photograph. Jezero comparison uses the repository's current built-world documentation and atlas generator.